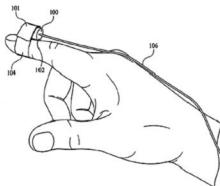


Technology News

Fingertip 'eye' to aid the blind

A small camera mounted on a fingertip could provide blind and partially sighted people a new way to interact with the world. Developed by George Stetten, a bioengineer at the University of Pittsburgh, this tiny video camera sits at the end of a finger and connects to a portable computer that can analyse the footage and flag up objects of interest.

The camera mounting could also vibrate to provide physical feedback from the computer to signal it has seen something significant. A warning could be given when the camera spots a nearby obstacle, like a wall or the edge of a table, and it could even trace out the shape to guide the user.



A small camera mounted on a fingertip could provide blind and partially sighted people a new way to interact with the world, a new patent application says

Image processing makes it possible for the system to recognise very specific objects. Amongst other things, the computer could recognise light switches or other controls from a distance. The user could be steered towards a switch using physical feedback, or given the option to activate it remotely.

Climbing snakebots

Snake-like robots may soon be lending construction workers, soldiers and



Snake-like robot, HyDRAS-Ascent prototype, has been designed to roll up poles, scaffolding or more complicated obstacles and may soon be lending construction workers and others a flexible hand

doctors a flexible hand. These flexible robots can climb scaffolding to inspect high-rise construction sites, bend at will to defuse roadside bombs or roll down your esophagus to peer at your digestive tract.

Dennis Hong, director of the Robotics & Mechanisms Laboratory (RoMeLa) at Virginia Tech, said most snake-inspired robots have been engineered for sidewinding or other ground-based locomotion. Likewise, his lab's HyDRAS-Ascent prototype (or Hyper-redundant Discrete Robotic Articulated Serpentine) can unfold and move around on the ground.

But the big idea behind the 1m long snakebot, which came to Hong, was to use flexible universal joints to let the mechanical snake roll up poles, scaffolds and other structures.

"The quick big picture application is reducing human injuries—and ultimately death—at construction sites," said Gabriel Goldman, a graduate student in Hong's lab who has worked

extensively on the award-winning project.

Tiny motors to power surgical bots

Surgical procedures could soon be performed with tiny robots. Motors about twice the thickness of a human hair could be used to power mini robots to fly around inside the body.

Conventional electric motors do not perform well as these are scaled down in size. To address that, research in recent years has seen the use of so-called piezoelectric materials. These are typically crystals that expand and contract when a voltage is applied to them.

That makes 'linear motors,' which simply move back and forth at high frequency. But for real motion within the body, the micro-motors need to be able to rotate.

The new research leverages the same approach, by transforming the linear motion of tiny piezoelectric motors into rotation. That is accomplished by coupling the motors to a structure with a helix-shaped cut in it. Because the structure is held in place along the helical groove, a push at one end is turned into a rotation.

Robot assistant has surgical precision

Using a robot to operate a camera in gall bladder operations is as safe as working with a human, British researchers said in an analysis that underscores the effectiveness of robot technology in surgery.

While robots may not be ready to take over, they offer vast potential to do things such as allowing doctors to perform emergency operations without having to wait for a human assistant, Kurinchi Gurusamy of the Royal Free Hospital in London and colleagues